

Probably

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1 Expected Distance Between Two Points of a Square

Let points P, Q be sampled uniformly at random from the unit square. Compute the expected value of the length of line segment PQ .

1. Note that $\int \sec^3 \theta d\theta = \frac{1}{2}(\sec \theta \tan \theta + \log |\sec \theta + \tan \theta|) + C$. A similar formula can be used to compute $\int \sec^5 \theta d\theta$.

2. Let $X_1, X_2, Y_1, Y_2 \in I$ be iid uniformly distributed rvs. Note that $X_1 - X_2, Y_1 - Y_2$ have triangle distributions and we wish to compute

$$4 \int_0^1 \int_0^1 (1-z)(1-w) \sqrt{z^2 + w^2} dz dw.$$

Then split up the integral. The most nontrivial is the integral $\int_0^1 \int_0^1 \sqrt{z^2 + w^2} dz dw$, which is computed by change of coordinates to polar.

2 More Continuous Expectations

The expected distance from a point picked uniformly at random from a unit disc to the center (origin) is $2/3$.

Suppose $X_1, X_2, \dots, X_n \in I$ are iid rv. Let $Y = \min_{i \in [n]} X_i$. $P(Y \geq y) = \prod P(X_i \geq y) = P(X \geq y)^n = (1 - F_X(y))^n$. $f_Y(y) = n(1 - F_X(y))^{n-1} f_X(y)$.

$$\begin{aligned} E[Y] &= \int_0^1 y f_Y(y) dy = \int_0^1 y \cdot n(1 - F_X(y))^{n-1} f_X(y) dy \\ &= -y(1 - F_X(y))^n \Big|_0^1 + \int_0^1 (1 - F_X(y))^n dy \\ &= \int_0^1 (1 - F_X(y))^n dy = \int_0^1 P(Y \geq y) dy, \end{aligned}$$

giving the tail sum formula for continuous rvs.

3 Dice Games

Suppose we roll a dice repeatedly and add to our score if the roll is new and subtract otherwise, and that we can stop any time. What is the optimal strategy?

Denote by $\sum S$ the sum of the elements of S for finite $S \subset \mathbb{R}$. I claim that we should continue rolling if $\sum S \leq 10$ and stop otherwise.

Denote by $E(S)$ the expected value given optimal play if the elements of S have been rolled before. $E([6]) = 0$ since we should stop. Suppose the result holds for cardinality $n \leq 6$ and I wish to show for $n-1$. Let $|S| = n-1$ such that $\sum S \geq 11$. Then if we roll, $E(S) = -\frac{1}{6} \sum S + \frac{|S|}{6} E(S) + \frac{1}{6} \sum S^c + \frac{1}{6} \sum_{x \in S^c} E(S \cup x) = \frac{7}{2} - \frac{1}{3} \sum S + \frac{|S|}{6} E(S)$, so $\frac{6-|S|}{6} E(S) = \frac{7}{2} - \frac{1}{3} \sum S < 0 \rightarrow E(S) < 0$ (since $|S| \neq 6$). Hence, we should not play, so $E(S) = 0$.

For $\sum S \leq 10$, we have from the same formula (using nonnegativity of $E(A)$), $\frac{6-|S|}{6} E(S) \geq \frac{7}{2} - \sum S \geq 0 \rightarrow E(S) \geq 0$ so we should play (as expected), and we're done.